

EMBEDDED SYSTEMS & IoT DEVICE DESIGN

Task 3: Smart Automatic Light Control and Alert System

Electronics & Communication Engineering

Maincraft Technologies Training Program

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Project Report

**Smart Automatic Light Control and Alert System
Using LDR Sensor and Buzzer**

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1 Objective

The objective of this project is to design and simulate a Smart Automatic Light Control and Alert System using an LDR (Light Dependent Resistor), a Piezo Buzzer, and an Arduino UNO[cite: 1]. The system automatically controls a dual-LED configuration (Green and Red) and triggers an audio alarm based on ambient light intensity thresholds[cite: 1].

2 Introduction

Automation plays an important role in modern embedded systems[cite: 2]. Smart lighting and automated safety alert systems help reduce energy consumption by automatically switching lights ON or OFF according to surrounding light conditions while offering real-time monitoring[cite: 1, 2].

This project demonstrates the interfacing of an LDR sensor and a Piezo Buzzer with Arduino UNO to create an integrated automatic light control and warning environment[cite: 1, 2].

3 Components Used

S.No.	Component	Quantity
1	Arduino UNO	1
2	LDR (Light Dependent Resistor)	1
3	Red LED	1
4	Green LED	1
5	Piezo Buzzer	1
6	10k Ω Resistor	1
7	220 Ω Resistors	3
8	Breadboard	1
9	Jumper Wires	As required

4 Theory of LDR Sensor

An LDR (Light Dependent Resistor) is a sensor whose resistance changes according to the intensity of light falling on it[cite: 2].

4.1 Characteristics

- High resistance in darkness[cite: 2].
- Low resistance in bright light[cite: 2].
- Analog output suitable for Arduino ADC[cite: 2].
- Low-cost and easy to use[cite: 2].

4.2 Working

The LDR forms a voltage divider circuit with a resistor[cite: 2]. The output voltage varies according to light intensity and is read by Arduino through Analog Pin A0[cite: 2].

5 Circuit Design

The LDR and $10k\Omega$ resistor form a voltage divider circuit[cite: 2]. The junction point is connected to Analog Pin A0 of Arduino UNO[cite: 2].

- The Red LED is connected to Digital Pin 2 through a 220Ω resistor[cite: 2].
- The Green LED is connected to Digital Pin 3 through a 220Ω resistor.
- The Piezo Buzzer is connected to Digital Pin 4 through a protective resistance network[cite: 1].

6 Working Principle

The Arduino continuously reads the analog value from the LDR sensor[cite: 2].

- **Low Light Condition** $\rightarrow$ Red LED ON, Green LED OFF, Buzzer ON (Alert status)[cite: 1].
- **High Light Condition** $\rightarrow$ Red LED OFF, Green LED ON, Buzzer OFF (Normal status)[cite: 1].

6.1 Threshold Logic:

- LDR Value $< 500 \rightarrow$ Red LED ON & Buzzer ON (Dark Environment)[cite: 1, 2]
- LDR Value $\geq 500 \rightarrow$ Green LED ON & Buzzer OFF (Bright Environment)[cite: 1, 2]

7 Arduino Program

```
1 // Smart Light Control and Alert System using LDR and Buzzer
2 int ldrPin = A0;
3 int redLedPin = 2;
4 int greenLedPin = 3;
5 int buzzerPin = 4;
6
7 void setup() {
8     pinMode(redLedPin, OUTPUT);
9     pinMode(greenLedPin, OUTPUT);
10    pinMode(buzzerPin, OUTPUT);
11    Serial.begin(9600);
12 }
13
14 void loop() {
15     int ldrValue = analogRead(ldrPin);
16
17     if (ldrValue < 500) {
18         digitalWrite(redLedPin, HIGH);
19         digitalWrite(greenLedPin, LOW);
20         digitalWrite(buzzerPin, HIGH);
21         Serial.println("Dark Environment Detected");
22         Serial.print("LDR Value = ");
23         Serial.println(ldrValue);
24     }
25     else {
26         digitalWrite(redLedPin, LOW);
27         digitalWrite(greenLedPin, HIGH);
28         digitalWrite(buzzerPin, LOW);
29         Serial.print("LDR Value = ");
30         Serial.println(ldrValue);
31         Serial.println("Bright Environment");
32     }
33     delay(1000);
34 }
```

8 Simulation Results

8.1 Complete Circuit

The circuit baseline diagram below features organized component alignment across the breadboard. Jumper paths link the operational channels directly back to the Arduino board framework.

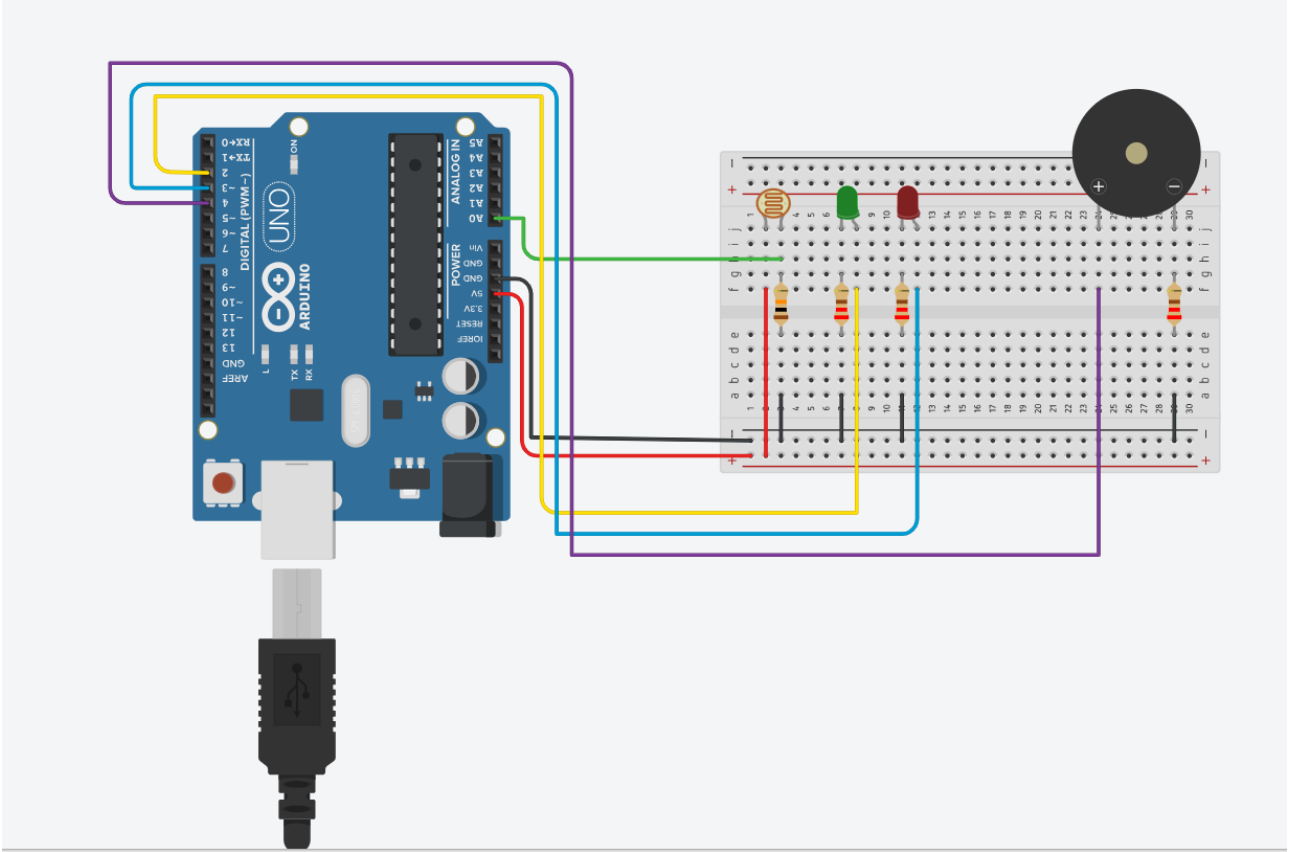


Figure 1: Complete Circuit Diagram on Tinkercad Platform

8.2 Low Light Condition

When the environment shifts towards zero-lux values, the system logic triggers active responses automatically.

- **Observation:** The LDR outputs a level of **54**. The Green line switches off, the Red LED fires active visual illumination, and the Piezo Buzzer operates active acoustic alert waves[cite: 1].

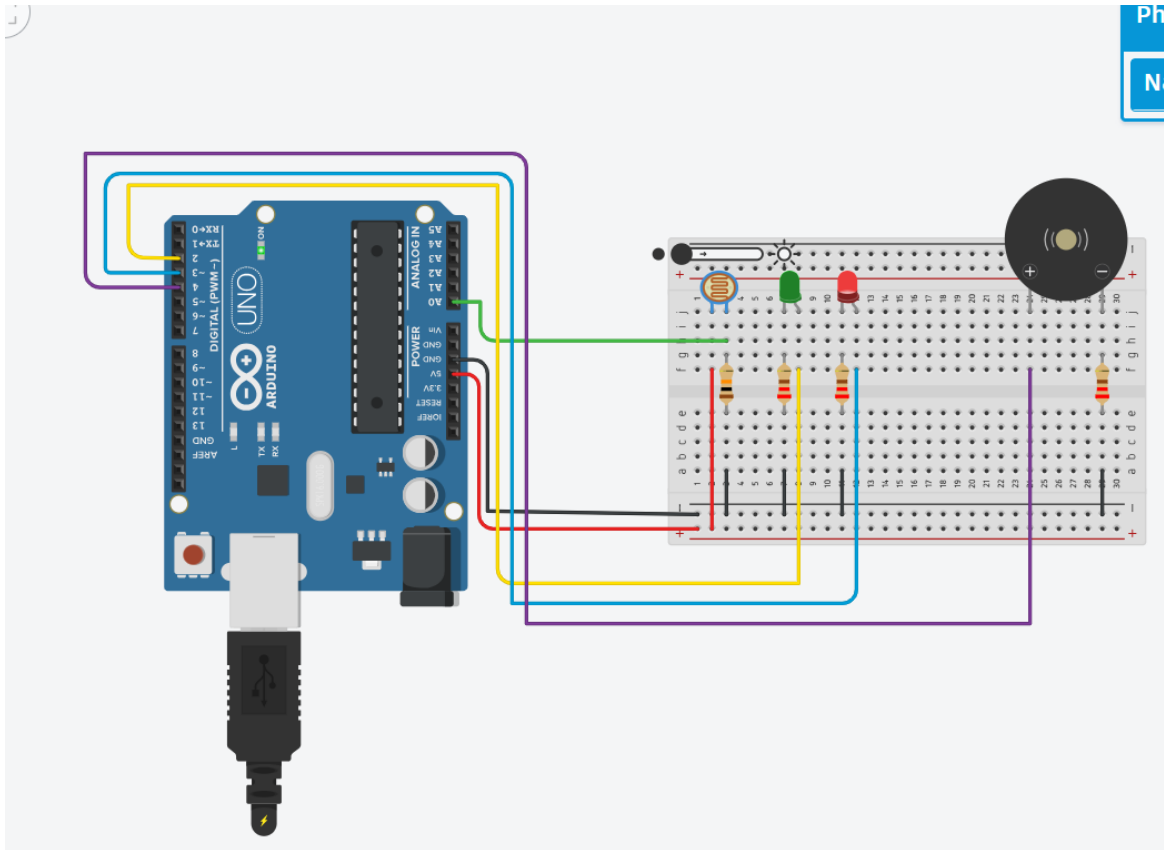


Figure 2: LDR Value = 54, Red LED ON and Piezo Buzzer Active

8.3 High Light Condition

Optimal external illumination updates the operational variables back to a standby pattern.

- **Observation:** The monitored LDR value logs an updated parameter of **716**. The controller responds immediately: the Green LED triggers ON, and the warning channels drop low[cite: 1, 2].

8.4 Serial Monitor Output

Continuous real-time evaluations from the terminal screen capture the automated system changes perfectly:

```
--- Live Serial Monitor Output Log ---
Dark Environment Detected
LDR Value = 54
Dark Environment Detected
LDR Value = 54
Dark Environment Detected
LDR Value = 716
Bright Environment
```

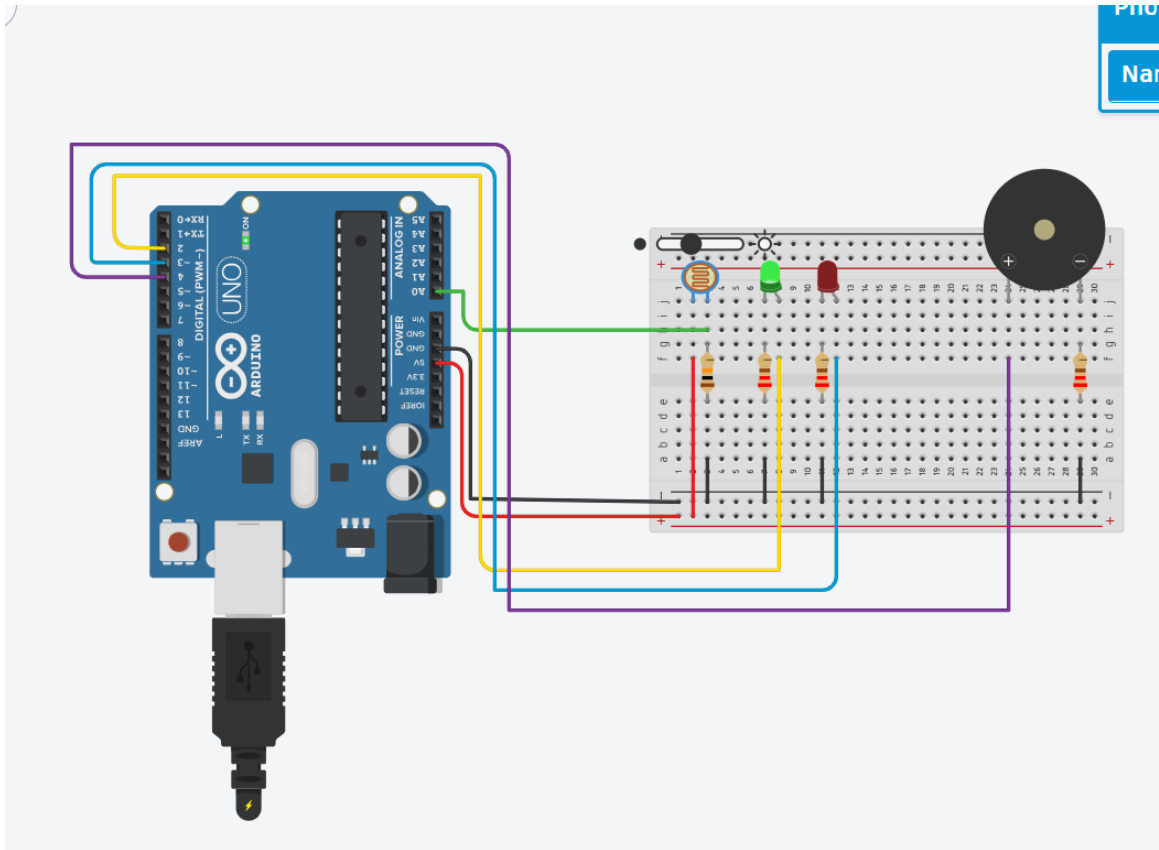


Figure 3: LDR Value = 716, Green LED ON and Alarm System Deactivated

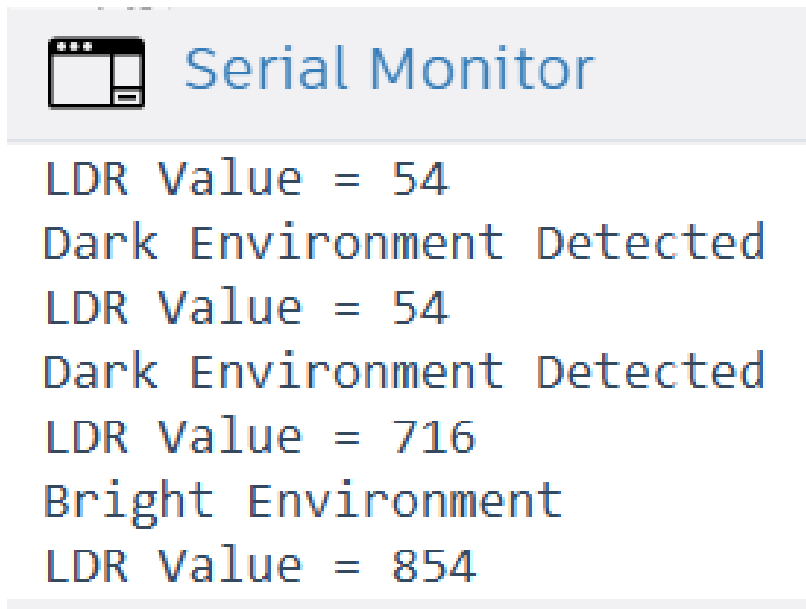


Figure 4: Real-Time LDR Values Displayed on Serial Monitor Terminal

9 Applications

- Automatic Street Lighting Systems[cite: 2]
- Smart Home Automation and Intrusive Alerts[cite: 1, 2]

- Energy Saving Lighting Systems[cite: 2]
- Industrial Safety and Blackout Alarms[cite: 1, 2]
- IoT-Based Smart Buildings[cite: 2]

10 Advantages

- Low Cost Implementation[cite: 2]
- Easy to Design and Maintain[cite: 2]
- Automatic Multi-Sensory Operation (Audio + Visual)[cite: 1, 2]
- Energy Efficient[cite: 2]
- Suitable for Smart City Safety Applications[cite: 2]

11 Conclusion

The Smart Automatic Light Control and Alert System was successfully designed and simulated using Arduino UNO, an LDR sensor, and a Piezo Buzzer[cite: 1]. The project successfully demonstrates sensor interfacing, analog signal processing, peripheral alert control, and embedded programming concepts[cite: 2].

12 References

1. Arduino Official Documentation
2. Tinkercad Circuits Simulator
3. LDR Sensor Datasheet
4. Arduino UNO Datasheet
5. Embedded Systems and IoT Training Material[cite: 2]